LAB SHEET

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B. TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LAB SHEET – 4

Title: Implementation of **User Creation, Deletion, and Privilege Management in SQL**

Problem Statement: Design and execute SQL commands to **create users, manage their access privileges, and remove users** in a database system. This assignment focuses on **database security and access control mechanisms**.

Database security ensures that only authorized users can access or modify data. MySQL provides commands such as:

- CREATE USER → to create new users
- GRANT → to give permissions
- REVOKE → to remove permissions
- DROP USER → to delete users Privileges include:
- SELECT (read data)
- INSERT (add data)
- UPDATE (modify data)
- DELETE (remove data)

Step 1: Create Database and Tables

```
CREATE DATABASE LAB_SHEET_4;
USE LAB_SHEET_4;
```

Explanation:

- Creates a database named LAB_SHEET_4
- Selects it for further operations

Create Tables

```
CREATE TABLE Student (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT
);
```

Explanation:
Creates a Student table where:

- id is unique
- name stores student name
- age stores age

```
CREATE TABLE Course (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(50)
);
```

Explanation:
Creates a Course table to store course details.

Part A: User Creation

Step 2: Create user1

```
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'pass123';
```

Explanation:

- Creates a user named user1
- 'localhost' means user can log in from same system
- IDENTIFIED BY sets password

Step 3: Create user2

```
CREATE USER 'user2'@'localhost' IDENTIFIED BY 'pass124';
```

Explanation:
Creates another user user2 with login credentials. **Part B:**

Granting Privileges

Step 4: Grant SELECT and INSERT to user1

```
GRANT SELECT, INSERT ON LAB_SHEET_4.Student TO 'user1'@'localhost';
```

Explanation:

- SELECT → allows viewing data

- INSERT → allows adding data
- Access is limited to Student table only

Step 5: Grant ALL privileges to user2

```
GRANT ALL PRIVILEGES ON LAB_SHEET_4.Course TO 'user2'@'localhost';
```

Explanation:

- Gives full access (SELECT, INSERT, UPDATE, DELETE, etc.)
- Only on Course table

Step 6: WITH GRANT OPTION

```
GRANT SELECT, INSERT ON LAB_SHEET_4.Student TO 'user1'@'localhost' WITH GRANT OPTION;
```

Explanation:

- Allows user1 to **give permissions to other users**
- This is called **privilege delegation**

Part C: Revoking Privileges

Step 7: Revoke INSERT from user1

```
REVOKE INSERT ON LAB_SHEET_4.Student FROM 'user1'@'localhost';
```

Explanation:

- Removes only INSERT permission
- user1 can now only read data

Step 8: Revoke ALL from user2

```
REVOKE ALL PRIVILEGES ON LAB_SHEET_4.Course FROM 'user2'@'localhost';
```

Explanation:

- Removes all permissions
- user2 cannot access Course table anymore

Part D: User Deletion

Step 9: Drop user2

```sql
DROP USER 'user2'@'localhost';
```

Explanation:

- Permanently deletes the user
- All privileges are removed

Step 10: Show all users

```sql
SELECT user, host FROM mysql.user;
```

Output:

Result Grid | Filter Rows:

user	host
dept_user	localhost
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost
root	localhost
user1	localhost

Explanation:
Displays all users present in MySQL system.

Step 11: Show privileges of user1

```sql
SHOW GRANTS FOR 'user1'@'localhost';
```

Output:

Result Grid | Filter Rows: | Ex

Grants for user1@localhost
GRANT USAGE ON *.* TO \`user1\`@\`localhost\`
GRANT SELECT ON \`lab_sheet_4\`.\`student\` T...

Explanation:
Displays all permissions assigned to user1.